A study of Wavelet based Classification Algorithms of Multi-channel EEG signals

Karthik Uppuluri

Cullen College of Engineering, University of Houston, TX

***Abstract*— Electroencephalography serves as tool for analysis of brain signals. These studies of EEG signal forms an important branch of science in understanding different diseases associated with brain and understand the functionality of the brain. EEG signal classification is very much required in differentiating different states of the brain and helps in identifying different brain related disorders like seizures, coma, schizophrenia etc. This paper discusses the efficacy of different wavelet features, feature reduction methodologies and classification paradigms for multi-channel EEG signal classification. We demonstrated that for our dataset Haar and Symlet serve as better wavelet features, linear discriminant analysis serves as better feature reduction technique and Maximum-likelihood classifier as better classifier. We also demonstrated that ICA serves as good de-noising technique on contrary to wavelet thresholding.**

Keywords: EEG; Wavelets; Wavelet-Denoising; PCA; FLDA; Maximum-Likelihood classifier; SVM

I. INTRODUCTION AND LITERATURE

EEG is an acronym for Electroencephalography. EEG signals are spontaneous electrical activity of brain over a small period of time, usually 20-40 minutes obtained from multiple electrodes placed on the scalp[1]. Each neuron is considered as a source of electric charge and the EEG signal therefore reflects the summation of the synchronous activity of millions of neurons. Study of EEG plays a major role in the analysis of epileptic seizures and helps to differentiate it from non-epileptic seizures like fainting, movement disorders and migraine variants. Brain death is usually confirmed by analyzing EEG signal of that patient.

Usually the EEG signals are categorized into five types based on their frequency namely, Alpha waves (7.5-14 Hz), Beta waves (14-40 Hz), Gamma waves (above 40 Hz), Theta waves (4-7.5 Hz), Delta waves (0.5-4 Hz)[2]. These waves' in-turn helps in the analysis of mental condition of the patient. EEG signals are very sensitive to noise; the noise is mainly due to movement of different parts of the body like eyes, head neck during the EEG recording. Sometimes it may also be because of the electrodes. These noises which occur in EEG signals are called artifacts. For a proper analysis of these signals it is always better to remove them before proceeding to further analysis.

Different feature extraction techniques have been proposed in the literature for EEG signal classification. Most of them include wavelet features, 6th order Auto-regressive coefficients obtained by Burgs method, Adaptive-autoregressive coefficients, and amplitudes of EEG signals, Band powers, Power Spectral Density values, time-frequency features and inverse model-based features[3]. Pfurtscheller et.al reported a classification accuracy of 83.6% using Band power as a feature and Linear Discriminant analysis as a classifier[4]. Most of them reported higher accuracies when they used autoregressive or adaptive autoregressive coefficients. Penny et al. and Pfurtscheller et al. reported accuracies up to 85% when they used adaptive auto regressive coefficients along with Multi-Layer Perceptron as their classifier[5][6]. Nazari et al. used Haar wavelet basis for feature extraction and reported accuracies of 88.75% and 80.87 % when Multi-layer perceptron is used as a classifier for two different datasets[7]. Most of wavelet based feature extractions in the literature suggest Haar as a good wavelet basis.

Feature reduction techniques in literature can be briefly classified as linear feature reduction techniques and non-linear feature reduction techniques[8]. One of the most common linear techniques is Principal Component analysis or Hotelling transform[9]. It is a linear transformation from a set in high dimensional space to a low dimensional space. However non-linear feature reduction methods have said to be giving better accuracy than linear feature reduction techniques[10]. Classifiers can also be divided into linear and nonlinear classifiers. Linear classifiers are the discriminant algorithms which use linear functions to distinguish the classes[3]. Two main linear classifiers are LDA and SVM. In this paper we tried to study the efficacy of Radial Basis Kernel based SVM also called as Gaussian SVM. Hidden Markov models, Bayesian classifiers come under nonlinear classifiers. However these classifiers are not as wide spread as linear classifiers[3]. Maximum-Likelihood classifier is rarely discussed in the literature with respect to EEG signal classification. The reason is it's a naïve classifier as it assumes the data in each class to be a normal distribution which is not true with respect to EEG signals. We have analyzed the efficacy of this classifier in this experiment. The data set we used is affected by muscular artifacts. Muscular artifacts are really difficult to remove due to their high amplitude, large frequency range and variable topographical distribution[12]. Literature proves that ICA serves as a better method for Denoising of muscular artifacts[12]. In this paper we also compared the Denoising results with wavelet based soft thresholding method [13].

II. DATA AND METHODS

A. Data

We used a seventeen class dance data set as our experimental dataset. The data was collected from a

professional dancer as she was executing four basic movements in sequence namely time, space, flow, and weight with each movement having two polarities. There are sixteen movements with one neutral state giving rise to seventeen classes namely 0-neutral, 1-think: quick time, 2-do: quick time, 3-think: sustained time, 4-do: sustained time, 5-think: direct space, 6-do: direct space, 7-think: indirect space, 8-do: indirect space, 9-think: bound flow, 10-do: bound flow, 11-think: free flow, 12-do: free flow, 13-think: light weight, 14-do: light weight, 15-think: strong weight, 16-do: strong weight.

B. Methods

We extracted five different wavelet features, tested two different feature reduction techniques and two different classification algorithms. The data is said to be corrupted by muscle artifacts. SO, in addition we have denoised the data using Independent Component Analysis and wavelet based thresholding. We used Coiflet1, Coiflet2, Daubechies4, Haar and Symlet2 as our wavelet basis for feature extraction. Linear dimension reduction techniques like Principal Component Analysis and Fisher's Linear Discriminant Analysis have been used for feature reduction. Maximum-Likelihood classifier is initially used to test the efficacy of the wavelet features and feature reduction techniques.

After feature extraction we have feature matrices for each wavelet basis and for each scale. As the size of the feature matrices is huge we used PCA and FLDA.

B.1 PCA+ML

From feature matrices we have we chose some samples as our training set and some samples as our testing set. For example we have an n-d dimensional training set $X_{dxn} = \{ x_1, x_2, x_3, ...x_n \}$, m-d-dimensional testing samples $y_{dxm} = \{ y_1, y_2, y_3, ...y_m \}$. We initially subtract the mean which is the average across the dimension of the training/testing data from training and testing datasets. Calculate the co-variance matrix of the training data along with the Eigen vectors and Eigen values. Choose the eigenvectors with the highest eigenvalues as transformation matrix $W_{(d\times r)}$, where r is the number of eigenvectors chosen. We then multiply training and testing data by transformation matrix separately to obtain projected training and testing data

$$\begin{aligned} \tilde{x}_{rxn} &= W_{dxr}{}^{T}\, X_{dxn} = \{ \tilde{x}_1, \tilde{x}_2, \tilde{x}_3 .....\tilde{x}_n \} \\ \tilde{Y}_{rxn} &= W_{dxr}{}^{T}\, Y_{dxm} = \{ \tilde{y}_1, \tilde{y}_2, \tilde{y}_3 .....\tilde{y}_m \} \end{aligned} .$$

Assuming the distribution of the data is Gaussian we use Maximum-Likelihood to estimate mean and covariance of each class calculated from the projected training data. The estimated mean of each class is given by $\hat{\mu}_{i_{rx1}} = \frac{1}{l_i}\sum_{K=1}^{ll_i} \tilde{x}_k$ and the estimated covariance matrix is given by $\hat{\Sigma}_{i_{rxr}} = \frac{1}{l_i - 1}\sum_{k=1}^{l_i} (\tilde{x}_k - \hat{\mu}_i)(\tilde{x}_k - \hat{\mu}_i)^T$ where l_i is the number of samples in each class i. The discriminant function is then used to classify the testing data and the function is given by

$$g_i(\tilde{y}) = \ln p(\tilde{y}/c_i) + \ln P(c_i)$$

Where $p(c_i) = \frac{\text{samples in class i of training data}}{\text{total samples of training data}}$ and $p(x/c_i)$ is the multi-variate normal density function and c_i is the class label. Rewriting the formula

$$\begin{aligned} g_i(\tilde{y}) &= -\frac{1}{2}(\tilde{y} - \hat{\mu}_i)^T \hat{\Sigma}_I^{-1} (\tilde{y} - \hat{\mu}_i) \\ &\quad - \frac{r}{2}\ln 2\pi - \frac{1}{2}\ln|\hat{\Sigma}_i| + \ln P(c_i) \end{aligned}$$

Where r is the dimension of the data $\tilde{Y}_{rxm}$, $|\hat{\Sigma}_i|$ is the determinant of $\hat{\Sigma}_i$ .Assign $\tilde{y}$ to class w_i if $g_i(\tilde{y}) > g_j(\tilde{y})$ for all $j = i$

B.2 FLDA+ML

For linear discriminant analysis we initially calculate a within class scatter matrix $S_{w_{dxd}}$ and between class scatter $S_{b_{dxd}}$ for the training data. Check if the within class scatter matrix is of full rank, if it is directly compute the inverse or else compute the pseudo inverse of that matrix. Calculate the Eigen values of $S_w^{-1}S_b$ and sort them in descending order. Then chose the Eigen vectors with highest eigen values as the transformation matrix W_{dxr} where r is the number of eigen vectors chosen. And repeat the last three steps of B.1 from projecting training and testing data to classification.

B.3 ICA Denoising

For independent component analysis we used an algorithm called fastica [11]. The raw EEG signal is assumed to be a linear combination of actual signal, noise and effect of other signals reaching at different electrodes. FastICA gives an unmixing matrix which can differentiate the original independent signal from noise and other background sources. The denoised results have been shown in results section.

$$\begin{aligned} x(t) &= A\, s(t) \\ \hat{s}(t) &= W\, x(t) \end{aligned}$$

Where A is called mixing matrix, W is unmixing matrix and S is the independent signal.

B.4 Wavelet Thresholding

In wavelet thresholding based denoising an EEG signal is decomposed into low frequency and high frequency coefficients. At each level low frequency coefficents are thresholded using a soft threshold

$T = \sqrt{2\sigma^2 \frac{\log(n)}{n}}$ Where σ the standard deviation and n is the length of the sequence. In this experiment we went up to four levels to denoise the data and we use Coiflet as our wavelet basis. After level four the wave is reconstructed using the thresholded low pass coefficients and high pass coefficients and this will be our denoised signal.

III. RESULTS

As mentioned above we used a 17 class dance data set collected from a professional dancer to test the efficacies of five different wavelet features, two different feature reduction techniques and two different classification algorithms. In addition we have also attached the Denoising results obtained from Independent Component Analysis and wavelet based thresholding and compared the accuracy of classifier with de-noised data. We used Coiflet1, Coiflet2, Daubechies4, Haar and Symlet2 as our wavelet basis for feature extraction. We used fastica to obtain the unmixing matrix and with wavelet thresholding we used soft thresholding and went upto four levels of wavelet decomposition. For linear discriminant analysis we tried to check accuracy with dimension reduction factor varying from 2 to 16. We used 100 training examples per class with principal component analysis and 200 training examples per class with linear discriminant analysis. We also compared the accuracy with scales 1 and 2.

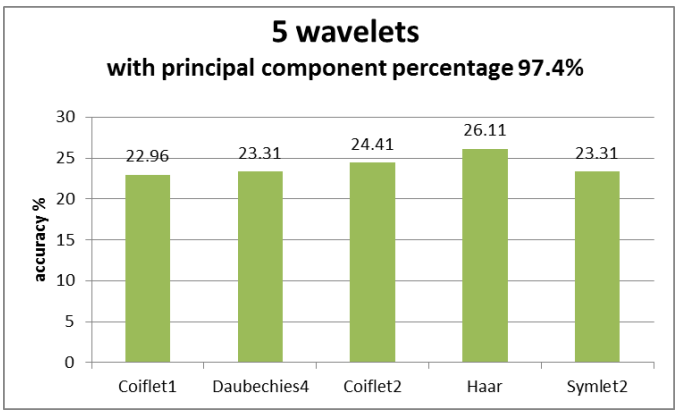

Figure.1 Comparison of accuracies of different wavelet features (scale1) with PCA and 97.4 % variance retained

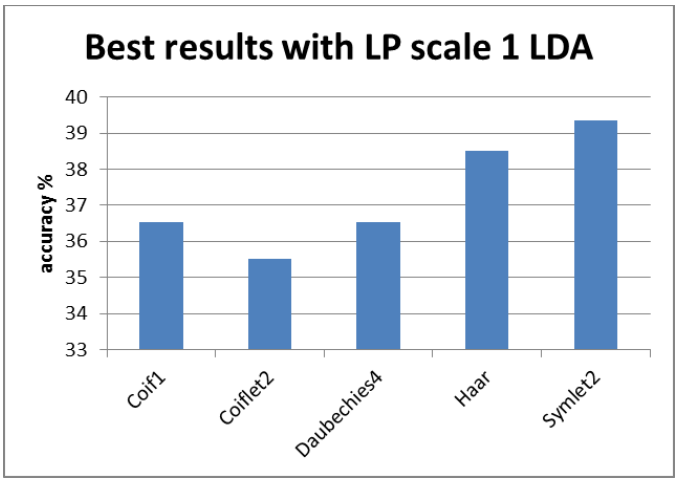

Figure2. Comparison of accuracies of different wavelet features (scale1) with linear discriminant analysis

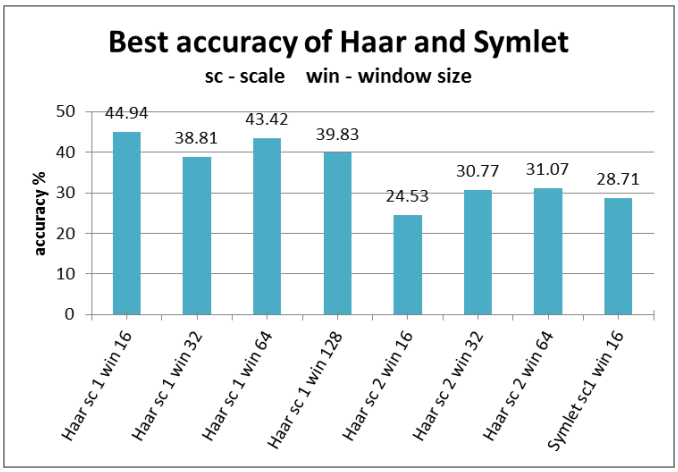

Figure.3 Comparison of accuracies of Haar and Symlet2 features with different scales and window sizes

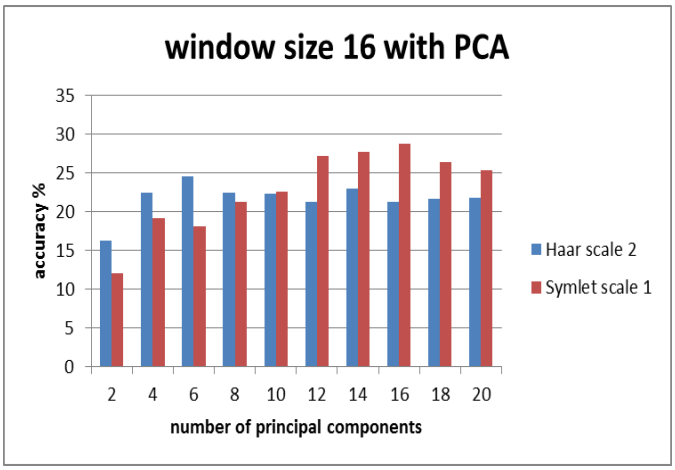

Figure.4.a Comparison of accuracy with number of principal components chosen

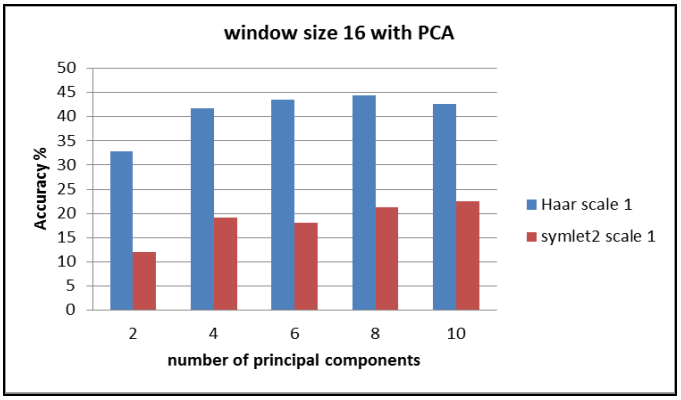

Figure4.b Comparison of accuracy with number of principal components chosen

In figure 1 we chose low pass coefficients of different waveletfeatures as our fetaure vectors for classification and we used principal component analysis as a feature reduction technique we took eigen values such that 97.4% of the variance is retained. We observed that accuracies are really low and comparitively Haar and symlet2 gave us better results. Even

though the acuurcay is low, one way in which we analysed it is, if we were to randomly guess to which class it belong our accuracy would be 5% so this result is better than that.

In figure 2 we changed our feature reduction technique to linear discriminant analysis, even though the dimension to which the data got reduced is less we noticed a significant improvement in classifier's performance. We got better accuracies with Haar and Symlet2 features which are around 38.5 and 39.5 %. As LDA considers both inter class and intra class scatter we can think of better accurate results than PCA which doesn't consider interclass scatter. Even though LDA is slower it continuously proved itself giving better accuracy throughout the experiment.

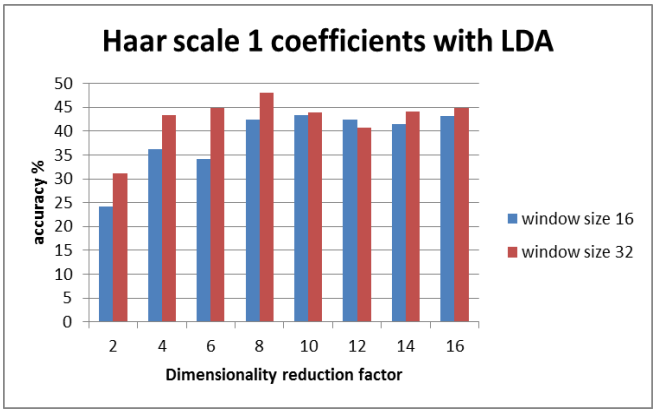

Figure5. Comparison of accuracy of Haar wavelet features with linear discriminant analysis

As Haar and Symlet2 were giving better results, we compared the accuracy of the classifier with Haar features at different scales and window sizes. In this case as the scale and window size increased the accuracy of the classifier decreased. The decrease in classifier accuracy with respect to window size can be explained this way. When we are taking larger and larger windows, one way we are assuming huge correlation between these points in the window, but on the contrary of the that correlation doesn't exist, in one way we are getting scarce of information thinking that the information with smaller windows is redundant. So the accuracy is getting decreased with larger window sizes. With respect to scale, I expected an increase in classification accuracy as the scale increases, because the scale increases we know more about the signal, as we are studying it at higher resolution so the features we get will be more efficient features, in turn classification accuracy should increase. But we didn't notice any increase in accuracy with respect to scale.

In figure 4.a and figure.4.b we can see the comparison of accuracies of Haar and Symlet 2 features at scale one and scale 2. We can notice that the accuracy of the classifier is better for scale1 Haar features when compared to scale 2 Haar and scale 1 Symlet2 features. So we considered that scale 1 Haar could give better accuracies than other features considered. Figure 5 gives the accuracy of Haar features with different window sizes and different dimension reduction factors. We got an accuracy of 48% with window size 32 and 44% with window size 16 and scale 1. So we noticed that with use of full coefficients instead of just low pass coefficients there is an increase in accuracy of the classifier

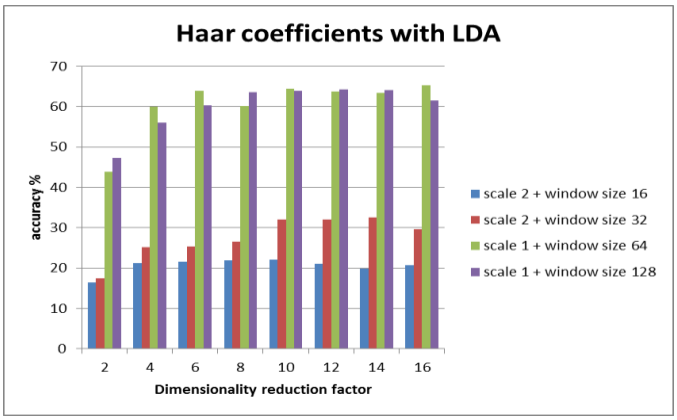

Figure6.a Combination of parameters which gave most accurate results with the dance dataset

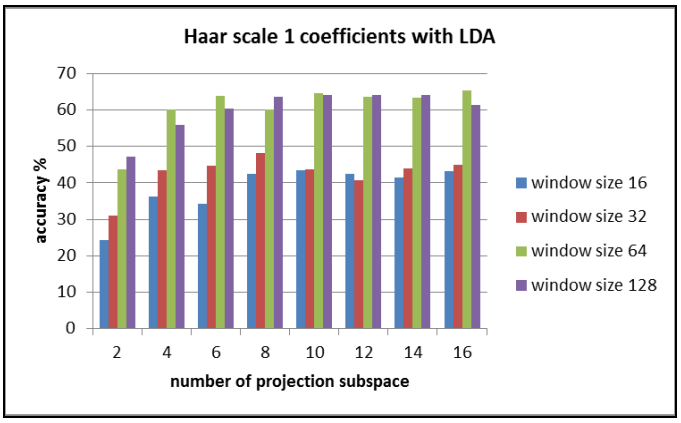

Figure6.b Comparison of accuracy of Haar coefficients at scale 1 and different window sizes with LDA

Figure 6.a describes the best results which got. It compares the Haar features at different scales, with different window sizes, LDA and the accuracy of the classifier. Figure6.b illustrates the results of Haar coeeficients at scale 1 and window sizes 16,32,64 and 128.

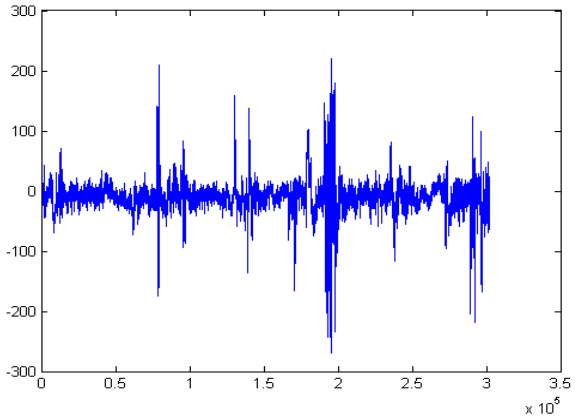

Figure.7. Original channel 49

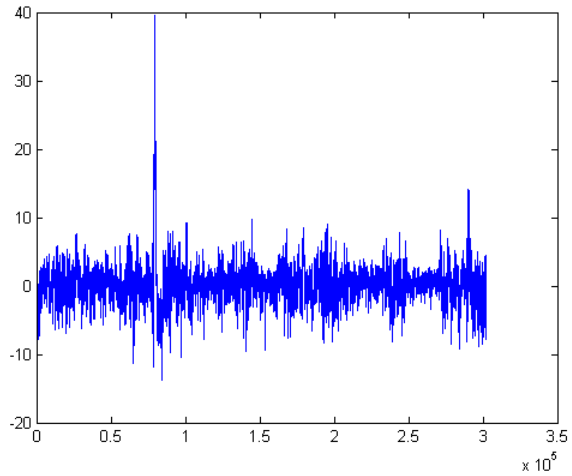

Figure.8 Channel 49 after unmixing using ICA

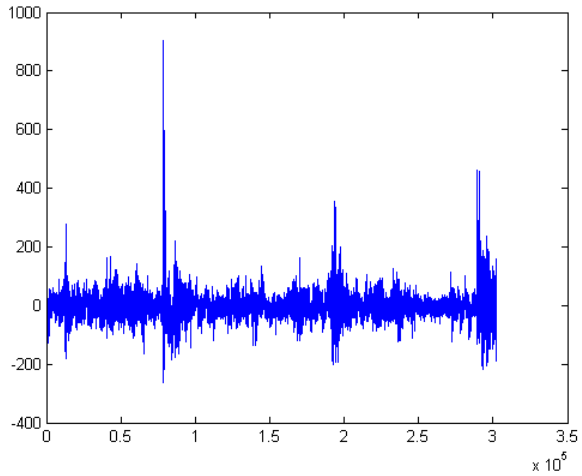

Figure.9 Original channel 64

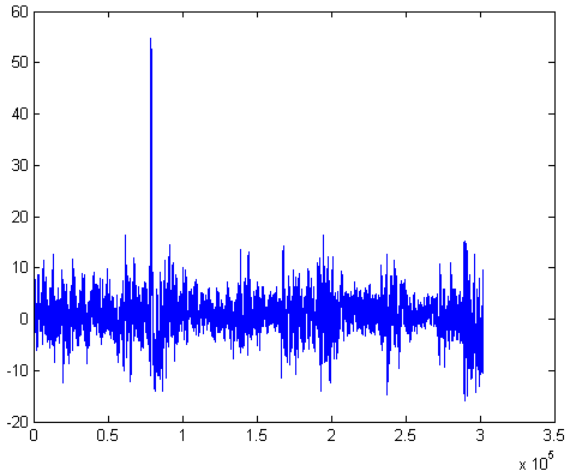

Figure.10 Channel 64 after Unmixing using ICA

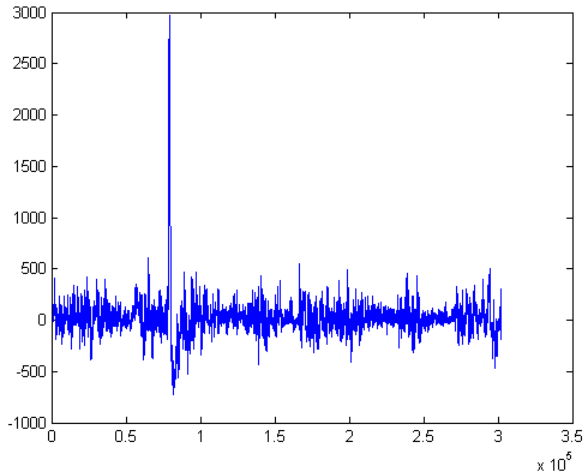

Figure.11 Original channel 33

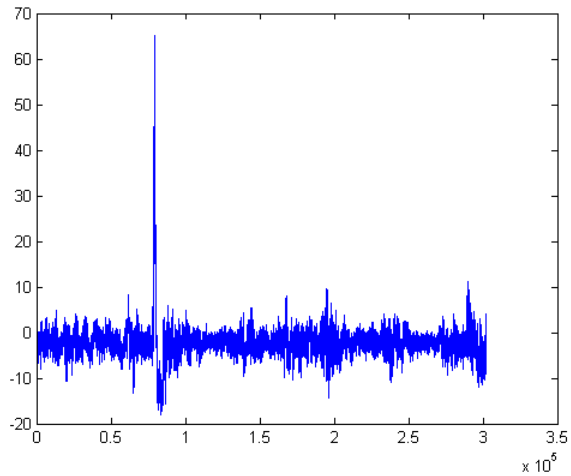

Figure12. Channel 33 after Unmixing using ICA

Figures 8,9,10,11,12 show the denoising results after unmixing the original signal from noise and other background sources.

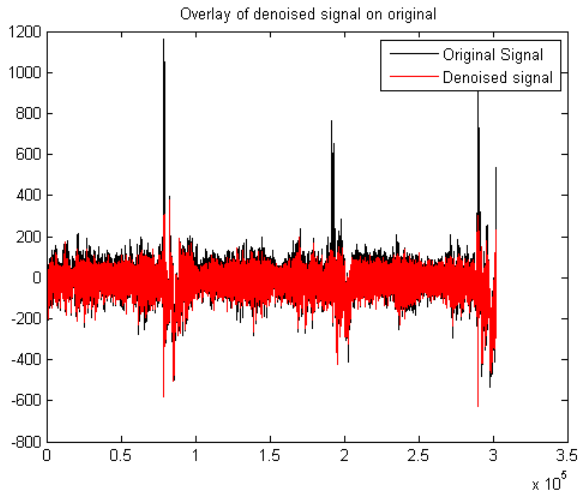

Figure.13 Overlay of original and denoised channel 17

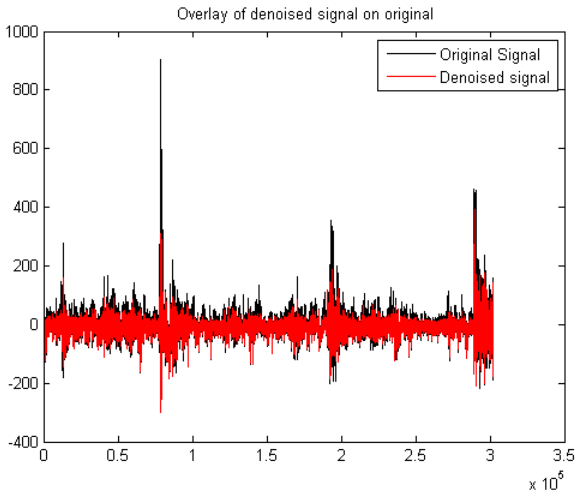

Figure.14 Overlay of original and denoised channel 64

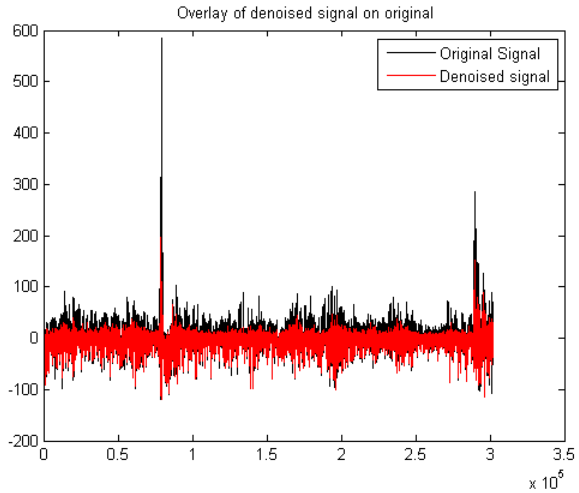

Figure.15 Overlay of original and denoised channel 56

Figures 13, 14, 15 represent the results obtained from wavelet thresholding. The red signal indicates the denoised signal. We have observed that the results obtained by wavelet thresholding are not good as there is a lot of information loss in the denoised signal. The results got converged as we reached level four. And the classification results with this data are really bad.

We compared the classification accuracies of the denoised data obtained from ICA.

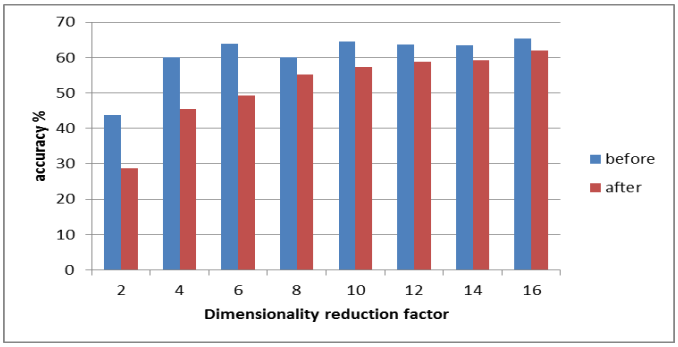

Figure.6 Accuracy after and before denoising

As we can see from this figure there is no significant change in the classification accuracy after denoising. We have also tried to classify the data with Haar features suing R.B.F SVM with different sigmas but none of them gave significant results which can be reported.

IV. CONCLUSION

Here by we conclude that we have achieved a classification accuracy of 65% by using Haar wavelet features at scale 1 and window size of 64 ms with Maximum Likelihood classifier. We noticed that for better classification accuracy Haar and Symlet2 serve as better wavelet basis. Linear Discriminant analysis can be preferred over Principal component analysis. We also noticed that ICA gives better denoising results as the structure of the EEG is maintained and we can clearly see that the amplitude values of the EEG signals stay below 60-70 microvolts as mentioned in the literature. Whereas with wavelet thresholding we can clearly see that most of the useful EEG signal information is lost. But there is no significant improvement in accuracy with denoised data. There is no improvement in classification accuracy with increase in scale. With increase the number of principal components and dimension reduction factor accuracy of the classifier increased. Instead of just using low pass coefficients both low pass and high pass coefficients should be used as features for better accuracies.

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